

EXP-12- EMBEDDED C PROGRAM TO MULTIPLY TWO 16-BIT BINARY NUMBERS

CODE:

```
#include <reg51.h>
```

```
void main(void)
```

```
{
```

```
    unsigned int num1, num2;
```

```
    unsigned long int product;
```

```
    num1 = 0x2222;
```

```
    num2 = 0xBBBB;
```

```
    product = (unsigned long int)num1 * num2;
```

```
    P0 = product & 0xFF;
```

```
    P1 = (product & 0xFF00) >> 8;
```

```
    P2 = (product & 0xFF0000) >> 16;
```

```
    P3 = (product & 0xFF000000) >> 24;
```

```
    while(1);
```

```
}
```

CIRCUIT DIAGRAM:

The screenshot displays a software development environment with three main panels:

- Registers Panel:** Shows the state of various registers. For example, `eax` contains `0x00000000`, `ecx` contains `0x00000000`, and `edx` contains `0x00000000`.
- Assembly Panel:** Displays assembly code with comments. The code includes instructions like `mov eax, 0x00000000`, `mov ecx, 0x00000000`, and `mov edx, 0x00000000`. It also shows a `while` loop structure.
- Circuit Diagram Panel:** Shows a logic diagram with four flip-flops labeled `Flip-Flop 0`, `Flip-Flop 1`, `Flip-Flop 2`, and `Flip-Flop 3`. Each flip-flop has inputs `Q0`, `Q1`, `Q2`, and `Q3` and outputs `Q0`, `Q1`, `Q2`, and `Q3`.
- Variable Table Panel:** A table listing variables and their values.

Name	Location/Value	Type
<code>main</code>	<code>0x00000000</code>	int
<code>mainC</code>	<code>0x00000000</code>	int
<code>product</code>	<code>0x00000000</code>	int